

Adjusting G by gain diagonal matrix to customize the objective function

$$G = \text{diag} \left(\begin{bmatrix} E_{G_1,t}^2 & E_{G_1}^1 \\ E_{G_2,t}^2 & E_{G_2}^1 \\ \vdots & \vdots \\ E_{G_q,t}^2 & E_{G_q}^1 \end{bmatrix} \begin{bmatrix} t_{\text{start}} \\ \mathbf{t} \\ t_{\text{end}} \end{bmatrix} \right) \Rightarrow G' = \begin{bmatrix} k_{G_1} & 0 & \cdots & 0 \\ 0 & k_{G_2} & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & k_{G_q} \end{bmatrix} \text{diag} \left(\begin{bmatrix} E_{G_1,t}^2 & E_{G_1}^1 \\ E_{G_2,t}^2 & E_{G_2}^1 \\ \vdots & \vdots \\ E_{G_q,t}^2 & E_{G_q}^1 \end{bmatrix} \begin{bmatrix} t_{\text{start}} \\ \mathbf{t} \\ t_{\text{end}} \end{bmatrix} \right)$$